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Project Conception
LinkedU Digital Platform

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1 Introduction

This project is carried out within the framework of an academic internship at LinkedU, an international education consultancy specialized in guiding students through their study-abroad journey. The internship focuses on the design and development of a web-based platform intended to digitalize and centralize the student guidance process offered by LinkedU.

Unlike a traditional informational website, the proposed solution is a multi-user interactive platform that integrates authentication, role-based access, real-time communication, progress tracking, and evaluation modules. The platform aims to improve the interaction between students and educational agents by providing a structured environment that supports academic orientation, preparation of application dossiers, interview preparation, document verification, and follow-up until admission confirmation.

This project allows the application of theoretical knowledge in web development, database management, and software engineering while addressing real operational needs within the company.

2 Company Presentation

Company Name: LinkedU

2.1 Company Overview

LinkedU is an international education consultancy company that specializes in assisting students who wish to pursue higher education abroad. The company provides structured academic guidance and personalized support throughout the entire study-abroad process. LinkedU operates as an intermediary between students and foreign educational institutions, helping candidates make informed academic choices and successfully complete their application procedures.

2.2 Mission and Objectives

The mission of LinkedU is to simplify and secure the study-abroad journey for students by offering professional academic orientation and continuous guidance. The company aims to:

- Help students identify suitable academic programs and destinations.
- Ensure the quality and completeness of application dossiers.
- Prepare students for interviews and institutional requirements.
- Monitor student progress until admission confirmation.

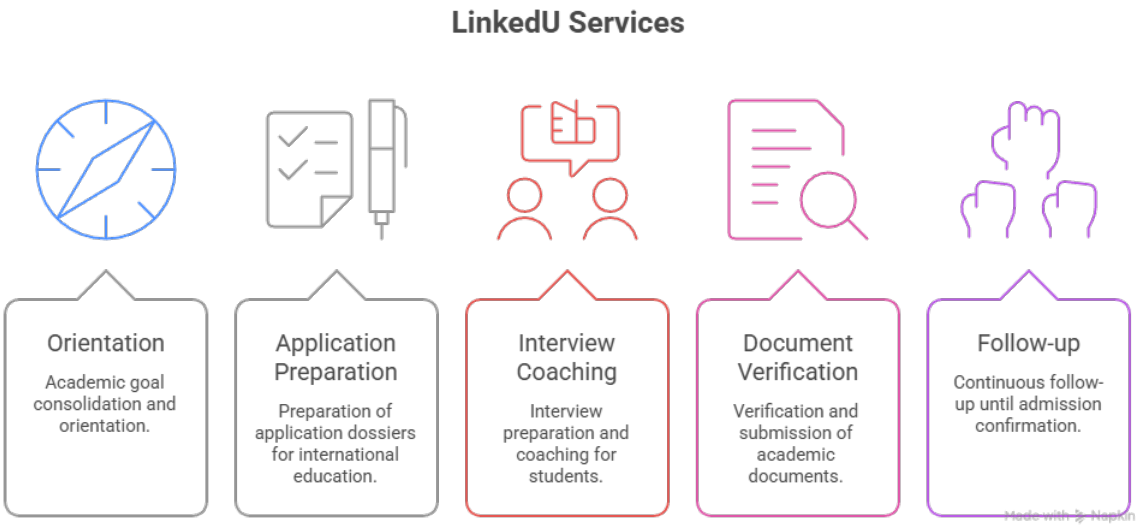
Visual summary of the LinkedU digital platform concept.

2.3 Services Provided by LinkedU

LinkedU provides a range of services related to international education, including:

- Orientation and consolidation of academic goals.
- Preparation of application dossiers.
- Interview preparation and coaching.
- Verification and submission of academic documents.
- Continuous follow-up until admission confirmation.

These services require continuous interaction, document tracking, and status monitoring, which justifies the need for a dedicated digital platform.



Services-focused visual for the LinkedU platform workflow.

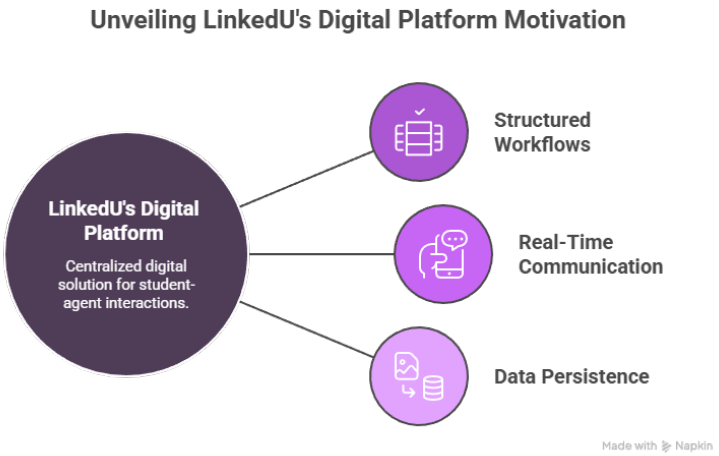
2.4 Target Users

LinkedU's services target several categories of users:

- **Students (Clients):** individuals seeking guidance and support for studying abroad.
- **Educational Agents:** staff members responsible for guiding and assisting students through each stage of the process.
- **Administrators:** personnel responsible for supervising operations, managing users, and monitoring performance.

2.5 Motivation for the Project

Due to the growing number of students and the complexity of application procedures, LinkedU requires a centralized digital solution to manage interactions, track progress, and ensure service quality. The development of a web-based platform addresses these needs by providing structured workflows, real-time communication, and data persistence, thereby improving efficiency and transparency for both students and agents.



3 Project Context and Problem Statement

3.1 Current Situation

Currently, the student guidance process at LinkedU relies on a combination of manual communication methods and basic digital tools such as email, messaging applications, and static web pages. Although the existing website provides general information about the company and its services, it does not support advanced interaction, personalized follow-up, or structured monitoring of student progress.

3.2 Identified Problems

This situation presents several limitations:

- Lack of a centralized system to manage student data.
- Difficulty in tracking the progress of each student across different guidance stages.
- Absence of role-based access for students, agents, and administrators.
- Communication scattered across multiple platforms without history or traceability.
- Limited visibility on agent performance and service quality.
- No automated assessment tools to evaluate student readiness (languages, orientation, etc.).

3.3 Consequences

These limitations can lead to:

- Delays in processing student applications.
- Inconsistent follow-up and guidance.
- Increased workload for agents.
- Reduced transparency for students regarding their application status.
- Difficulty for administrators to supervise activities and improve service quality.

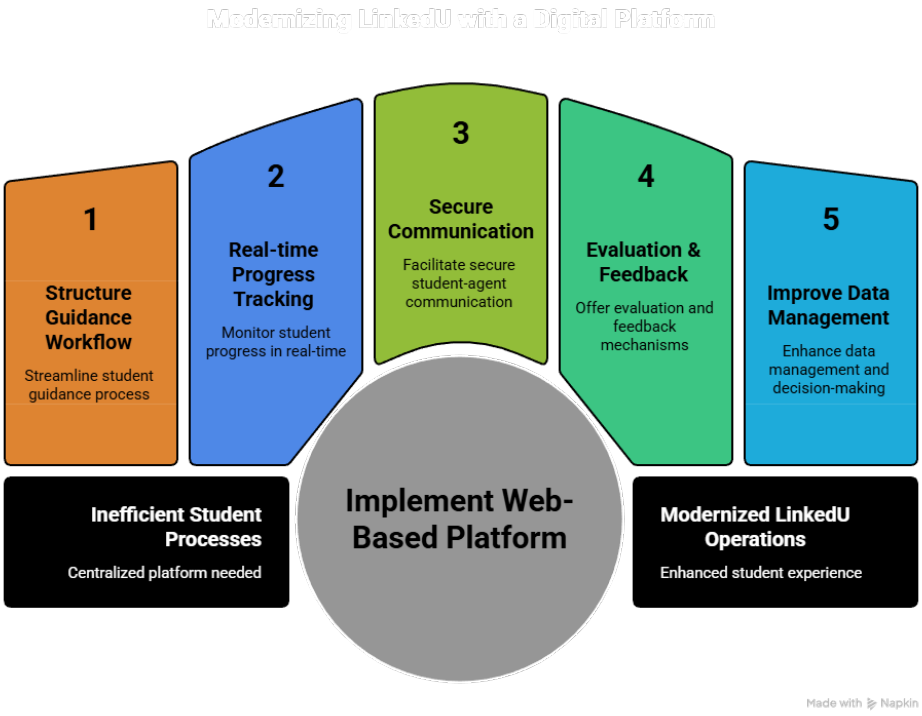
3.4 Need for a Digital Solution

To overcome these challenges, LinkedU requires a comprehensive web-based platform capable of centralizing all student-related processes. The proposed solution aims to:

- Structure the student guidance workflow.

- Provide real-time progress tracking.
- Facilitate secure communication between students and assigned agents.
- Offer evaluation and feedback mechanisms.
- Improve data management and decision-making.

This platform represents a strategic tool for modernizing LinkedU's operations.



4 Project Objectives

4.1 General Objective

The general objective of this project is to design and develop a secure, scalable, and interactive web platform that digitalizes the student guidance process at LinkedU by centralizing communication, progress tracking, and service management within a single system.

4.2 Specific Objectives

The specific objectives of the project are to:

- Implement a multi-user authentication system with role-based access (Student, Agent, Administrator).
- Allow students to register, log in, and manage their personal and academic profiles.
- Enable administrators to assign agents to students based on workload and specialization.
- Provide agents with tools to guide students through each stage of the study-abroad process.
- Develop a progress tracking system that records and displays the status of each student.
- Integrate a real-time communication module (chat) between students and their assigned agents.
- Implement a document management feature for uploading, verifying, and validating academic files.
- Introduce a quiz and evaluation module for language assessment and academic readiness.
- Collect student feedback and ratings for each agent to evaluate service quality.
- Provide administrators with dashboards and reports to monitor platform activity and performance.

5 Functional Requirements

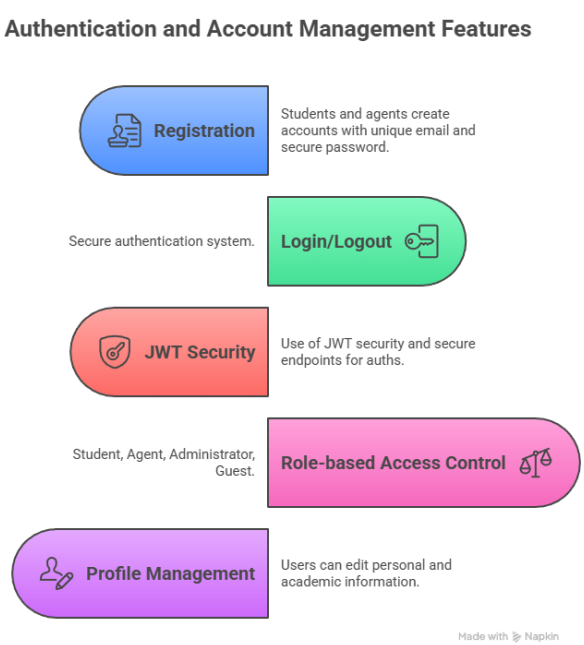
5.1 Authentication and Account Management

Purpose: Ensure secure access and role-based access control.

Features:

- **Registration:** Students and agents create accounts with a unique email and secure password.
- **Users with contract:** Can register as users and access dashboards.
- **Guest users:** Can register and access only public pages.
- **Login/Logout:** Secure authentication system.
- **Security:** Use of JWT (JSON Web Token) and secure endpoints for authentication.
- **Role-based Access Control:** Student, Agent, Administrator, and Guest roles.
- **Profile Management:** Users can edit personal and academic information depending on their role.

Actor(s) involved: Student, Agent, Administrator, Guest (all types of users).



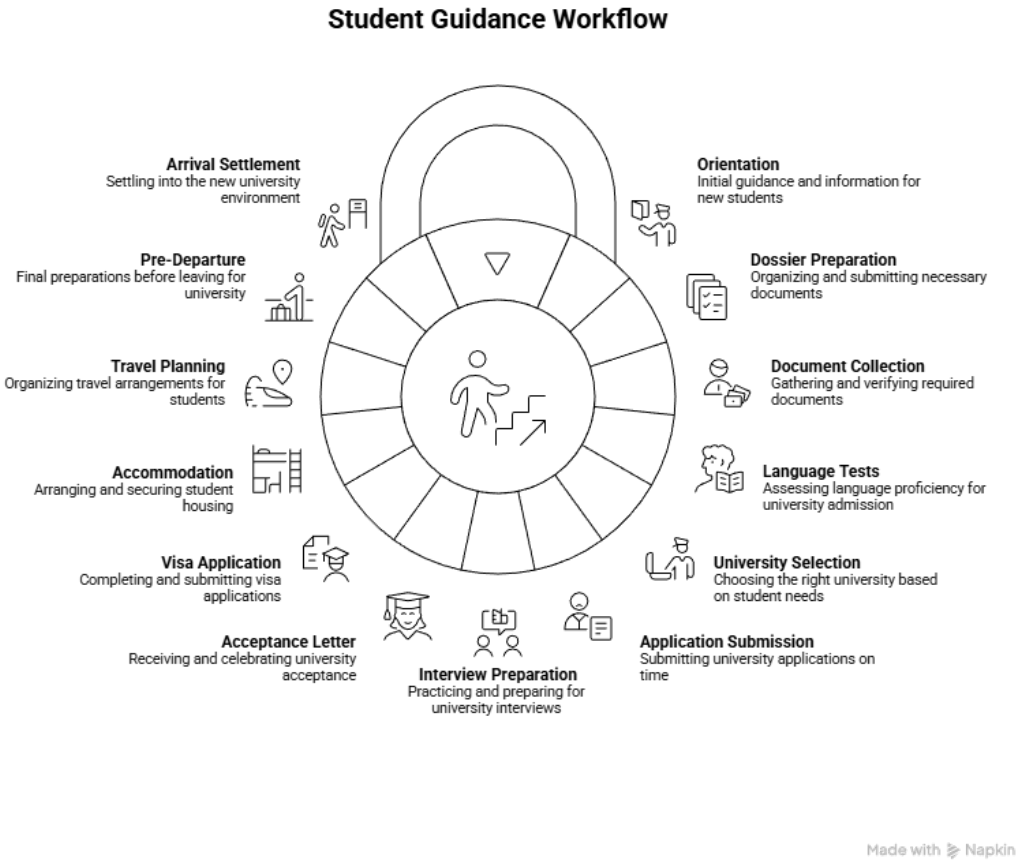
5.2 Student Progress Management

Purpose: Track and manage student guidance workflow.

Features:

- **Predefined workflow stages:**
  - ORIENTATION
  - DOSSIER_PREPARATION
  - DOCUMENT_COLLECTION
  - LANGUAGE_TESTS
  - UNIVERSITY_SELECTION
  - APPLICATION_SUBMISSION
  - INTERVIEW_PREPARATION
  - ACCEPTANCE_LETTER
  - VISA_APPLICATION
  - ACCOMMODATION
  - TRAVEL_PLANNING
  - PRE_DEPARTURE
  - ARRIVAL_SETTLEMENT
- Students can view progress per stage.
- Agents can update student progress.
- Administrators can monitor overall progress metrics.

Actor(s) involved: Student, Agent, Administrator


Student Guidance Workflow
Arrival Settlement
Settling into the new university environment
Orientation
Initial guidance and information for new students
Pre-Departure
Final preparations before leaving for university
Dossier Preparation
Organizing and submitting necessary documents
Travel Planning
Organizing travel arrangements for students
Document Collection
Gathering and verifying required documents
Accommodation
Arranging and securing student housing
Language Tests
Assessing language proficiency for university admission
Visa Application
Completing and submitting visa applications
University Selection
Choosing the right university based on student needs
Acceptance Letter
Receiving and celebrating university acceptance
Interview Preparation
Practicing and preparing for university interviews
Application Submission
Submitting university applications on time
Made with Napkin

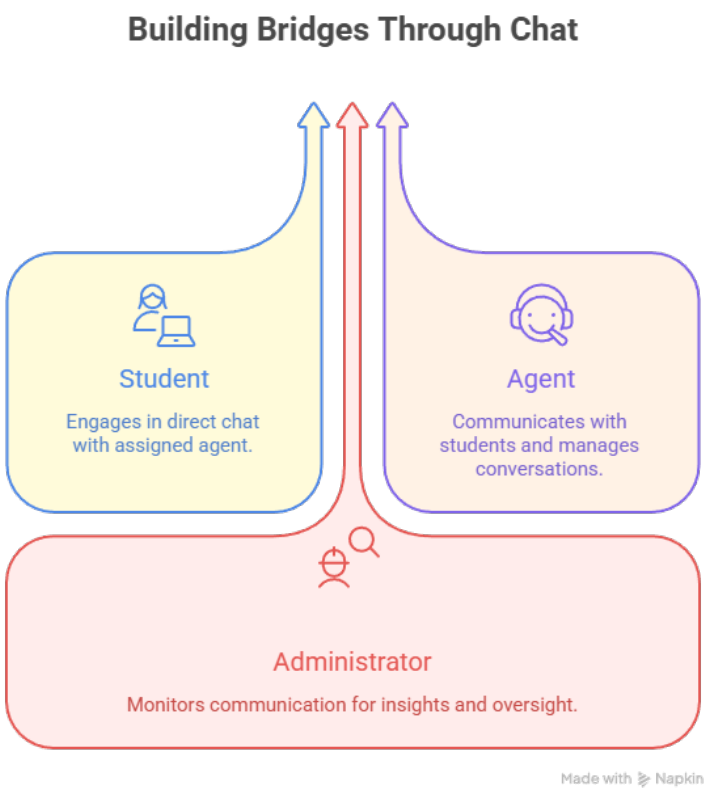
5.3 Communication Module (Chat)

Purpose: Enable real-time interaction.

Features:

- One-to-one chat between student and assigned agent.
- History of messages saved for reference.
- Notifications for new messages.
- Admin can monitor communication patterns (optional).

Actor(s) involved: Student, Agent, Administrator



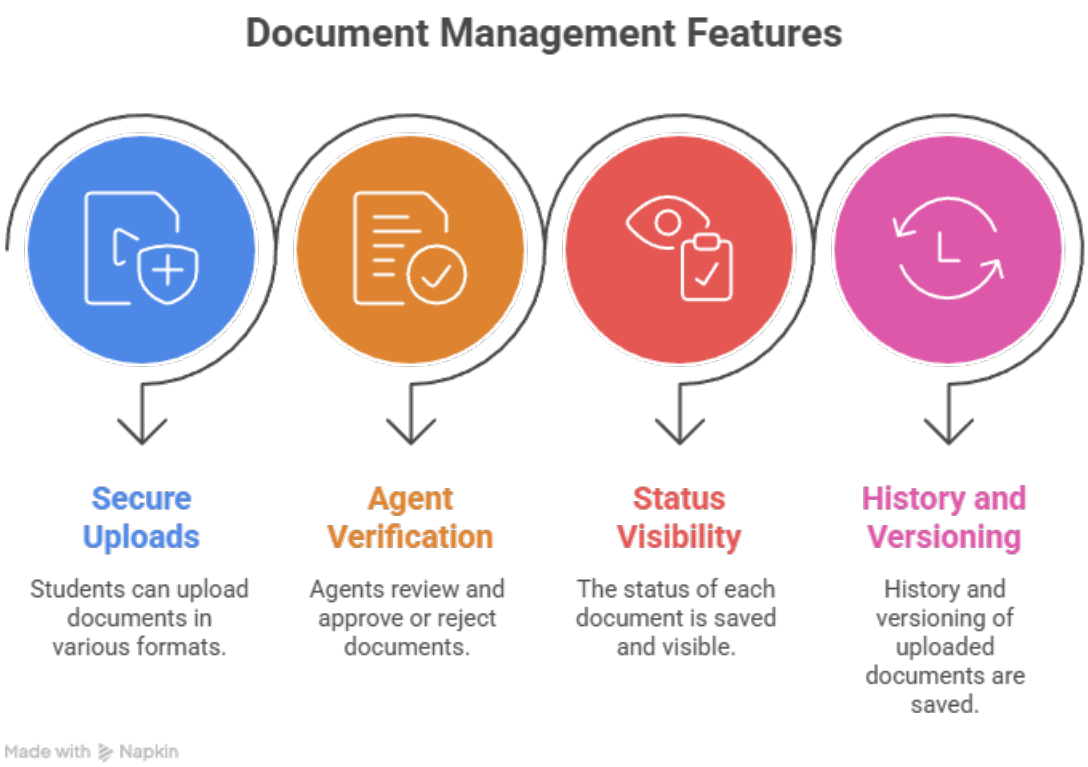
5.4 Document Management

Purpose: Manage academic documents efficiently.

Features:

- Students upload documents securely (PDF, JPG, DOCX).
- Agents verify and approve/reject documents.
- Status of each document is saved and visible.
- History and versioning of uploaded documents.

Actor(s) involved: Student, Agent, Administrator



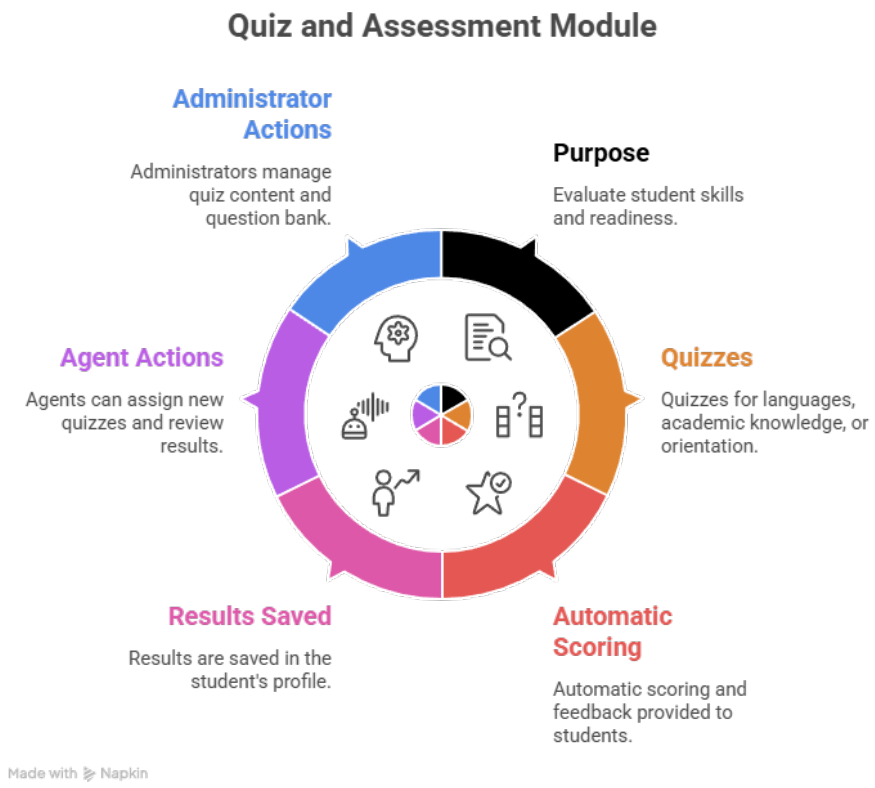
5.5 Quiz and Assessment Module

Purpose: Evaluate student skills and readiness.

Features:

- Quizzes for languages, academic knowledge, or orientation.
- Automatic scoring and feedback.
- Results saved in student profile.
- Agents can assign new quizzes and review results.
- Administrators manage quiz content and question bank.

Actor(s) involved: Student, Agent, Administrator



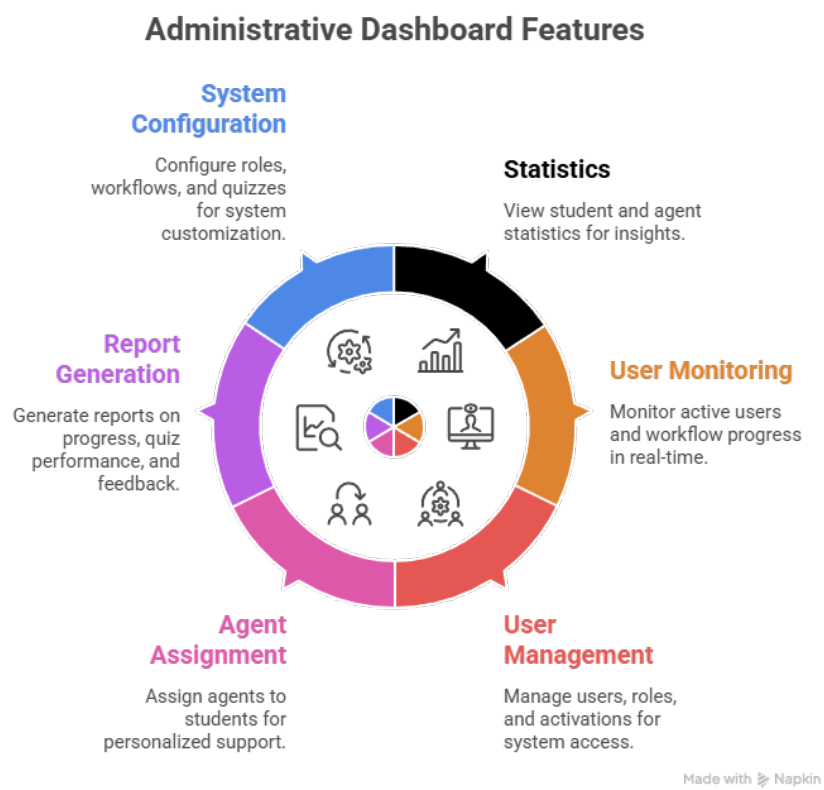
5.6 Administrative Dashboard and Reporting

Purpose: Give administrators control and overview.

Features:

- View student and agent statistics.
- Monitor active users and workflow progress.
- Manage users: roles and activations.
- Assign agents to students.
- Generate reports (progress, quiz performance, feedback).
- System configuration (roles, workflows, quizzes).

Actor(s) involved: Administrator



6 Non-Functional Requirements

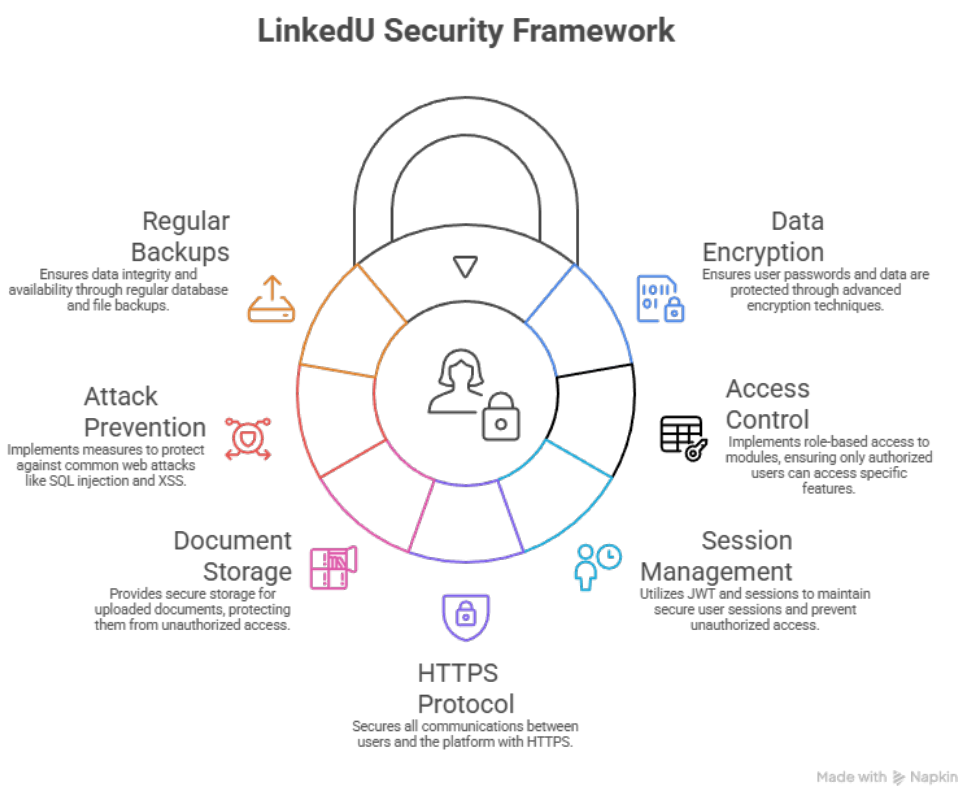
6.1 Security

Purpose: Protect user data and maintain trust.

Requirements:

- Passwords must be encrypted (e.g., bcrypt).
- Role-based access control for all modules.
- Usage of JSON Web Token (JWT) security and session management.
- HTTPS protocol for all communications.
- Secure storage of uploaded documents.
- Protection against SQL injection, XSS, and CSRF attacks.
- Regular backup of database and files.

Importance: Ensures that sensitive student and agent data is safe.



6.2 Performance

Purpose: Ensure smooth and fast operation.

Requirements:

- The system should support concurrent users without degradation.
- Page load time must be under 3 seconds for standard dashboards.
- Efficient database queries for progress tracking, quizzes, and messages.

Importance: Provides a good user experience and prevents delays.

6.3 Usability

Purpose: Make the platform intuitive and easy to use.

Requirements:

- Clean and modern interface for students, agents, and administrators.
- Responsive design for desktop, tablet, and mobile.
- Clear navigation menus and feedback messages.
- Forms with input validation and error handling.

Importance: Reduces training time and improves user satisfaction.

6.4 Scalability

Purpose: Allow future growth.

Requirements:

- The platform should allow adding new modules (e.g., new quizzes, feedback types) without major redesign.
- Database schema should support more students, agents, and documents.
- System architecture should allow deployment on larger servers or cloud services.

Importance: Ensures longevity and adaptability.

7 Technical Architecture

7.1 Overview of the Architecture

The platform follows a multi-tier web architecture consisting of:

1. **Presentation Layer (Front-end):** User interface, accessible via web browsers.
2. **Application Layer (Back-end):** Business logic, workflows, user authentication, communication, and document handling.
3. **Data Layer (Database):** Storage of student, agent, progress, document, quiz, and feedback information.

7.2 Front-End Layer

Purpose: Handle interaction with users (UI/UX).

Technologies: HTML5, CSS3, TypeScript, Font Awesome, Tailwind, Figma.

Framework: Angular for dynamic content and single-page application behavior.

Front-End Functional Characteristics:

- Responsive design for mobile and desktop.
- Input validation and user feedback.

Modules on Front-End:

AUTH MODELS

- `LoginPayload`
- `GuestRegisterPayload`
- `ContractRegisterPayload`
- `AuthResponse`
- `RegisterResponse`

USER / ROLES

- `UserRole` (type)
- `User`
- `AssignRolePayload`

AGENT MODELS

- `AssignedStudent`
- `DocumentResponseDTO`

STUDENT PROFILE MODELS

- `StudyLevel` (type)
- `CollegeType` (type)
- `Language`
- `StudentProfileDTO`
- `StudentProfileResponse`

PROFILE MODELS BASED ON ROLES

Shared

- `OnlineStatus` (type)

Guest

- `GuestProfileDTO`
- `GuestProfileResponse`

Agent

- `AgentProfileDTO`
- `AgentProfileResponse`

PRODUCT KEY

- `ProductKey`

DESTINATION MODELS

- `Destination`

QUIZ MODELS

- `Quiz`
- `QuizOption`

- `QuizQuestion`
- `QuizAttempt`
- `SubmitAnswerPayload`

CHAT MODELS

- `RoomType` (type)
- `MessageType` (type)
- `ParticipantRole` (type)
- `ChatRoom`
- `ChatParticipant`
- `ChatNotification`

DOCUMENT MODELS

- `DocumentStatus` (type)
- `DocumentType` (type)
- `StudentDocument`

PROGRESS MODELS

- `ProgressStage` (type)
- `ProgressStatus` (type)
- `Progress`

CHAT (MESSAGING)

- `ChatMessage`
- `UnreadCount`

TICKET MODELS

- `TicketStatus` (type)
- `Ticket`
- `CreateTicketDTO`
- `AcceptTicketDTO`
- `RejectTicketDTO`

7.3 Back-End Layer

Purpose: Business logic and processing.

Technologies:

- Java (Spring Boot) or Node.js.
- RESTful API to connect front-end and database.
- Authentication via JWT (JSON Web Tokens).
- Real-time chat with WebSockets.
- File management for document upload and storage.
- Business rules for progress stages, quiz scoring, and agent assignment.

7.4 System Modules and Functionalities

1. AUTH MODULE

- Authenticate user (email or username).
- Register guest.
- Register with contract (product key).
- Verify email.

2. USER MANAGEMENT (ADMIN)

- Get all users.
- Assign role to user.

3. DESTINATION MANAGEMENT (ADMIN)

- Create destination.
- Update destination.
- Delete destination.
- Get all destinations.
- Get destination by ID.

4. PRODUCT KEY MANAGEMENT (ADMIN)

- Generate product key.

- Get available keys.

5. AGENT PROFILE

- Create profile.
- Update profile.
- Get own profile.
- Get profile by user ID.
- Upload avatar.

6. STUDENT PROFILE

- Create profile.
- Update profile.
- Get profile.
- Upload avatar.

7. GUEST PROFILE

- Create profile.
- Update profile.
- Get profile.
- Get profile by user ID.
- Upload avatar.

8. QUIZ MANAGEMENT (TEACHER / AGENT)

- Create question.
- Create quiz.
- Get all quizzes.
- Get quiz by ID.
- Assign question to quiz.

9. STUDENT QUIZ FLOW

- Get quiz questions.
- Submit answers.

- Store attempt.

10. QUIZ ATTEMPTS

- Record attempt.
- Get attempts by student ID.

11. AGENT REVIEW SYSTEM

- Get pending attempts by agent.
- Evaluate answers.
- Assign score and pass/fail status.

12. CHAT SYSTEM

- Send message.
- Get conversation between users.
- Mark message as seen.
- Get unread messages by user.

13. DOCUMENT MANAGEMENT

- Upload CV.
- Upload passport.
- Upload ID card.
- Get documents by student ID.
- Verify document.

14. PROGRESS TRACKING

- Create progress stage.
- Get progress by student ID.
- Update progress status.
- Delete progress record.

15. TICKET SYSTEM (STUDENT AND AGENT)

Student Functionalities

- Create ticket.
- Get own tickets.

Agent Functionalities

- Get all tickets.
- Accept ticket.
- Reject ticket.
- Respond to ticket.

16. STUDENT MANAGEMENT

- Assign agent to student.
- Get students by agent ID.
- Get own assigned students.

8 API Endpoints Reference

8.1 Auth Module

Method	Endpoint	Description
POST	/api/auth/login	Login user
POST	/api/auth/register/guest	Register guest
POST	/api/auth/register/contract	Register with product key
GET	/api/auth/verify	Verify email

8.2 User Management

Method	Endpoint	Description
GET	/api/admin/users	Get all users
PUT	/api/admin/users/{userId}/role	Assign role

8.3 Destination Management

Method	Endpoint	Description
GET	/api/admin/destinations	Get all destinations
GET	/api/admin/destinations/{id}	Get destination by ID
POST	/api/admin/destinations	Create destination
PUT	/api/admin/destinations/{id}	Update destination
DELETE	/api/admin/destinations/{id}	Delete destination

8.4 Product Key Management

Method	Endpoint	Description
POST	/api/admin/product-keys	Create product key
GET	/api/admin/product-keys/available	Get unused keys

8.5 Agent Profile

Method	Endpoint	Description
POST	/api/agent-profile/create	Create agent profile

PUT	/api/agent-profile/update	Update agent profile
GET	/api/agent-profile/me	Get own profile
GET	/api/agent-profile/{userId}	Get profile by user ID
POST	/api/agent-profile/upload-avatar	Upload avatar

8.6 Student Profile

Method	Endpoint	Description
POST	/api/student-profile/create	Create student profile
PUT	/api/student-profile/update	Update student profile
GET	/api/student-profile/me	Get own profile
GET	/api/student-profile/{userId}	Get profile by user ID
POST	/api/student-profile/upload-avatar	Upload avatar

8.7 Guest Profile

Method	Endpoint	Description
POST	/api/guest-profile/create	Create guest profile
PUT	/api/guest-profile/update	Update guest profile
GET	/api/guest-profile/me	Get own profile
GET	/api/guest-profile/{userId}	Get profile by user ID
POST	/api/guest-profile/upload-avatar	Upload avatar

8.8 Quiz Management

Method	Endpoint	Description
POST	/api/questions	Create question
POST	/api/quizzes	Create quiz
GET	/api/quizzes	Get all quizzes
GET	/api/quizzes/{id}	Get quiz by ID
POST	/api/quiz-assignments	Assign question to quiz

8.9 Student Quiz Flow

Method	Endpoint	Description

GET	/api/student-quiz/quiz/{quizId}/questions	Get quiz questions
POST	/api/student-quiz/submit	Submit answers

8.10 Quiz Attempts

Method	Endpoint	Description
POST	/api/quiz-attempts	Record attempt
GET	/api/quiz-attempts/student/{studentId}	Get attempts by student

8.11 Agent Review System

Method	Endpoint	Description
GET	/api/agent-review/pending	Get pending attempts by agent
PUT	/api/agent-review/attempt/{attemptId}	Evaluate attempt

8.12 Chat System

Method	Endpoint	Description
POST	/api/chat	Send message
GET	/api/chat/conversation	Get conversation
PUT	/api/chat/{id}/seen	Mark as seen
GET	/api/chat/unread/{userId}	Get unread messages

8.13 Document Management

Method	Endpoint	Description
POST	/api/documents/cv	Upload CV
POST	/api/documents/passport	Upload passport
POST	/api/documents/id-card	Upload ID card
GET	/api/documents/student/{studentId}	Get documents by student
PUT	/api/documents/{documentId}/verify	Verify document

8.14 Progress Tracking

Method	Endpoint	Description
POST	/api/progress	Create progress stage
GET	/api/progress/student/{studentId}	Get progress by student
PUT	/api/progress/{id}/status/{status}	Update progress status
DELETE	/api/progress/{id}	Delete progress record

8.15 Ticket System

Method	Endpoint	Description
POST	/api/student/tickets	Student create ticket
GET	/api/student/tickets	Student get tickets
GET	/api/agent/tickets	Agent get all tickets
POST	/api/agent/tickets/{ticketId}/accept	Agent accept ticket
POST	/api/agent/tickets/{ticketId}/reject	Agent reject ticket
POST	/api/agent/tickets/{ticketId}/response	Agent respond to ticket

8.16 Student Management

Method	Endpoint	Description
POST	/api/students/assign-agent	Assign agent to student
GET	/api/students/agent/{agentId}/students	Get students by agent ID
GET	/api/students/my-students	Get own assigned students

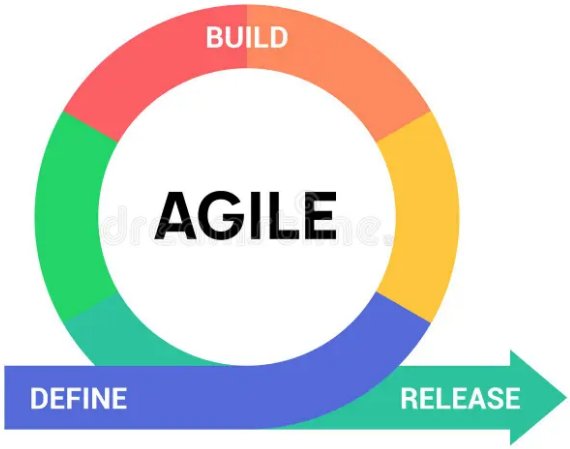
9 Development Methodology

9.1 Chosen Methodology: Scrum (Agile)

The project follows the Scrum Agile methodology, which is suitable for software development with frequent updates, changing requirements, and close collaboration.

Key Points of Scrum:

- Iterative and incremental development.
- Short cycles called sprints (e.g., 2 weeks).
- Regular meetings:
  - **Daily Standup:** track progress and obstacles.
  - **Sprint Planning:** select tasks for the next sprint.
  - **Sprint Review:** demonstrate completed features.
  - **Sprint Retrospective:** discuss improvements.
- Work organized in a Product Backlog and Sprint Backlog.



9.2 Roles in the Development Team

Role	**Responsibility**
Product Owner	Define requirements, prioritize backlog, ensure alignment with company goals.
Scrum Master	Facilitate meetings, remove obstacles, ensure Scrum practices are followed.
Development Team	Implement features, test modules, maintain documentation.
Developer	Mainly development, documentation, testing, and coordination with LinkedU staff.

10 Project Phases (Scrum Methodology)

10.1 Phase 1: Requirements Analysis and Backlog Creation

Sprint 0 – Setup and Planning

- Define product vision and scope.
- Create initial product backlog.
- Set up development environment.
- Define definition of done for user stories.
- Establish coding standards and guidelines.

Requirements Gathering

- Gather detailed requirements from LinkedU stakeholders.
- Identify functional requirements across all 16 modules.
- Identify non-functional requirements (security, performance, scalability).
- Define actors: ADMIN, AGENT, STUDENT, GUEST.

Backlog Refinement

- Define use cases and user stories for each actor.
- Create acceptance criteria for each story.
- Prioritize backlog items by business value.
- Estimate story points for each user story.
- Split large stories into smaller, testable increments.

Technical Assessment

- Review existing codebase and architecture.
- Identify technical dependencies between modules.
- Define API contracts and data flow.
- Document integration points.

10.2 Phase 2: System Design and Architecture

Architecture Planning

- Create system architecture diagrams.
- Design module interaction patterns.
- Define authentication and authorization flows (JWT, role-based).
- Plan database schema and relationships.

Database Design

- Create Entity Relationship Diagram (ERD).
- Design tables for all 16 modules.
- Define indexes, constraints, and foreign keys.
- Plan migration strategy.

API Design

- Define RESTful API endpoints for all modules.
- Create OpenAPI/Swagger specifications.
- Define request and response DTOs.
- Plan API versioning strategy.

Sprint Planning

- Break design work into sprints.
- Assign story points to design tasks.
- Create sprint goals and deliverables.

10.3 Phase 3: Implementation (Iterative Sprints)

Sprint Structure (2–3 weeks per sprint)

Each sprint includes:

- Sprint planning session.
- Daily stand-up meetings.
- Development tasks.
- Continuous testing.
- Sprint review.
- Sprint retrospective.

Sprint 1 – Core Infrastructure and Auth Module

- Set up back-end framework (Java Spring Boot / Node.js).
- Set up front-end framework (Angular).
- Configure database connection (MySQL/PostgreSQL).
- Implement JWT authentication.
- Build login and registration endpoints.
- Implement email verification.
- Create guest, student, and agent user types.
- Write unit tests for auth module.
- Sprint review and retrospective.

Sprint 2 – Profile Management

- Implement AGENT PROFILE module.
- Implement STUDENT PROFILE module.
- Implement GUEST PROFILE module.
- Add avatar upload functionality.
- Create profile update workflows.
- Write integration tests for profile modules.
- Fix bugs from previous sprint.
- Sprint review and retrospective.

Sprint 3 – Admin Management

- Implement USER MANAGEMENT (ADMIN) module.
- Implement DESTINATION MANAGEMENT module.
- Implement PRODUCT KEY MANAGEMENT module.
- Create admin dashboard interfaces.
- Implement role assignment workflows.
- Write tests for admin modules.
- Bug fixes and refinements.
- Sprint review and retrospective.

Sprint 4 – Quiz System

- Implement QUIZ MANAGEMENT module.
- Implement STUDENT QUIZ FLOW module.
- Implement QUIZ ATTEMPTS module.
- Implement AGENT REVIEW SYSTEM module.
- Create question and quiz creation interfaces.
- Build quiz taking and submission flow.
- Build agent evaluation interface.
- Write comprehensive tests.
- Sprint review and retrospective.

Sprint 5 – Communication Modules

- Implement CHAT SYSTEM module.
- Implement TICKET SYSTEM module.
- Add real-time messaging capabilities.
- Create ticket creation and response workflows.
- Implement unread message tracking.
- Write tests for communication modules.
- Bug fixes from previous sprints.
- Sprint review and retrospective.

Sprint 6 – Document and Progress Management

- Implement DOCUMENT MANAGEMENT module.
- Implement PROGRESS TRACKING module.
- Implement STUDENT MANAGEMENT module.
- Add document upload and verification.
- Create progress tracking dashboard.
- Build agent-student assignment workflows.
- Write integration tests.
- Sprint review and retrospective.

Sprint 7 – Front-end Integration and Polish

- Connect all front-end components to back-end APIs.
- Implement responsive design across all views.
- Add loading states and error handling.
- Optimize performance (lazy loading, caching).
- Cross-browser testing.
- Bug fixes and UI refinements.
- Sprint review and retrospective.

Sprint 8 – Security and Optimization

- Implement rate limiting and request validation.
- Add input sanitization and XSS protection.
- Configure CORS and security headers.
- Optimize database queries.
- Implement logging and monitoring.
- Security penetration testing.
- Sprint review and retrospective.

10.4 Phase 4: Testing (With Feedback and Repair Cycles)

Continuous Testing Within Sprints

- Unit testing during development (Jest, JUnit).
- Integration testing after each module completion.
- API testing using Postman/Newman.
- Front-end component testing.

Sprint Testing Activities

- Test-driven development (TDD) for critical features.
- Automated regression testing before sprint end.
- Peer code reviews and pair programming.
- Continuous integration pipeline testing.

User Acceptance Testing (UAT) Sprints

UAT Sprint 1 – Alpha Testing

- Deploy to staging environment.
- Internal testing by development team.
- Bug tracking and prioritization.
- Critical bug fixes within 24 hours.
- Minor bug fixes scheduled for next sprint.

UAT Sprint 2 – Beta Testing with LinkedU Staff

- Selected LinkedU staff test core modules.
- Collect feedback through issue tracking system.
- Weekly feedback review meetings.
- Prioritize feedback into product backlog.
- Immediate fixes for blocking issues.
- Scheduled fixes for non-critical issues.

UAT Sprint 3 – Full User Acceptance Testing

- All user roles (ADMIN, AGENT, STUDENT) test full system.
- Real-world scenarios and workflows.
- Performance and load testing.
- Security testing and vulnerability assessment.
- Feedback collection and analysis.

Testing Feedback Loop Process

Daily

- Bug reports reviewed and triaged.
- Critical bugs fixed within 24 hours.
- Testers verify fixes before closure.

Weekly

- Testing metrics reviewed (bug count, pass rate, coverage).

- Backlog reprioritized based on feedback.
- Sprint plan adjusted for high-priority fixes.

End of Each Testing Sprint

- Defect analysis meeting.
- Root cause identification.
- Process improvement for next sprint.
- Retest all fixed issues.

10.5 Phase 5: Deployment and Release

Pre-Deployment Sprint

- Final regression testing.
- Performance benchmarking.
- Security audit completion.
- Data migration testing.
- Rollback plan preparation.

Deployment Execution

- Deploy platform to production server or cloud (AWS, Azure, GCP).
- Configure database replication and backups.
- Set up SSL/TLS certificates.
- Configure domain and DNS settings.
- Implement CDN for static assets.

Production Configuration

- Set up monitoring and alerting (Prometheus, Grafana).
- Configure log aggregation (ELK stack).
- Implement automated backup schedule.
- Set up disaster recovery procedures.
- Configure auto-scaling policies.

Post-Deployment Sprint

- Smoke testing in production environment.
- Monitor error rates and performance.
- Address any deployment issues.
- Create hotfix procedure.

Training Delivery (Conducted in parallel with deployment)

- Train LinkedU administrators on system management.
- Train agents on platform usage.
- Create training videos for students.
- Conduct live training sessions.
- Provide quick reference guides.

10.6 Phase 6: Documentation and Handover

Technical Documentation

- System architecture document.
- Database schema documentation.
- API reference documentation (Swagger/OpenAPI).
- Deployment and operations guide.
- Security and compliance documentation.
- Backup and recovery procedures.

User Documentation

- Administrator manual.
- Agent user guide.
- Student user guide.
- Guest user guide.
- FAQ documents for each user type.
- Troubleshooting guides.

Handover Sprint

- Final code repository cleanup.

- Tag release version.
- Document known issues and workarounds.
- Transfer credentials and access securely.
- Schedule post-launch support handoff.

Post-Launch Support (First 30 Days)

- Dedicated support team availability.
- Daily health checks.
- Bug fix sprints as needed.
- Performance tuning based on real usage.
- Collect enhancement requests for next release.

10.7 Scrum Ceremonies Summary

Sprint Planning Sprint Planning is conducted every 2–3 weeks and typically lasts between 2 and 4 hours. Its purpose is to define the sprint goals, clarify priorities, and select the backlog items that will be implemented during the sprint.

Daily Stand-up The Daily Stand-up takes place every day and is time-boxed to 15 minutes. The team synchronizes progress, communicates completed work, and identifies blockers that require immediate attention.

Sprint Review Sprint Review is held at the end of each sprint for 1 to 2 hours. During this ceremony, the team demonstrates completed work to stakeholders and collects feedback to refine the product backlog.

Sprint Retrospective Sprint Retrospective is also organized at the end of each sprint and usually lasts 1 hour. It focuses on identifying process improvements, team collaboration enhancements, and practical actions for the next sprint.

Backlog Refinement Backlog Refinement is scheduled weekly for approximately 1 hour. The team prioritizes, clarifies, and estimates backlog items to ensure that upcoming sprint planning remains efficient and realistic.

Bug Triage Bug Triage is performed daily in 15-minute sessions. Reported issues are reviewed, prioritized, and assigned to appropriate team members based on impact and urgency.

10.8 Feedback and Repair Dynamics

Continuous Feedback Channels The platform uses multiple continuous feedback channels: an in-app feedback button for testers, weekly stakeholder demo sessions, a bug tracking system such as Jira or GitHub Issues, and customer support ticketing integration. Together, these channels ensure that functional, usability, and operational issues are quickly captured.

Priority Levels for Repairs Critical issues (system downtime, data loss, or security breaches) require a 1-hour response time and are fixed within 24 hours. High-priority issues (major feature failures blocking workflows) require a 4-hour response and are targeted for correction within 3 days. Medium-priority issues (minor non-blocking feature problems) require a response within 1 day and are addressed in the next sprint. Low-priority items (UI glitches and enhancement requests) receive a response within 1 week and are scheduled in the backlog.

Quality Gates Quality gates include minimum code coverage of 80%, zero critical security vulnerabilities, resolution of all critical bugs before sprint closure, 100% passing integration tests, and fulfillment of defined performance benchmarks.

10.9 Summary of Phases

Phase 1 Estimated duration: 2–3 weeks. Main deliverables include the product backlog, detailed user stories, and acceptance criteria.

Phase 2 Estimated duration: 2–3 weeks. Main deliverables include architecture diagrams, the database schema, and API specifications.

Phase 3 Estimated duration: 12–16 weeks. Main deliverable is a fully functional application delivered across 8 sprints.

Phase 4 Estimated duration: 4–6 weeks. Main deliverables include test reports, resolved defects, and UAT sign-off.

Phase 5 Estimated duration: 2–3 weeks. Main deliverables include live production deployment and trained staff.

Phase 6 Estimated duration: 1–2 weeks. Main deliverables include complete documentation and finalized handover.

11 Expected Results

After implementing the LinkedU platform through Scrum-based development with continuous testing, structured feedback, and repair cycles, the project is expected to deliver measurable operational, academic, and organizational improvements.

11.1 Centralized Student Management

All student information, academic progress, quiz results, documents, and communication history are centralized within a unified platform. Administrators can monitor student progress through real-time dashboards, while assigned agents and administrators access profiles according to role-based permissions. Historical data remains preserved and searchable across all 16 modules, and guest, student, and agent profiles are managed coherently within the same system.

11.2 Efficient Agent–Student Interaction

Each student is assigned to an agent, either automatically or manually, to ensure consistent guidance throughout orientation, applications, and follow-up steps. A real-time chat system supports immediate communication, while the ticketing module provides structured handling of complex issues. Agents can manage all assigned students from a single interface, and students can initiate conversations, track response status, and benefit from read receipts and unread notifications that strengthen accountability.

11.3 Progress Tracking

Agents can update each student's progress through predefined workflow stages such as document upload, quiz completion, and application submission. Students can view progress in real time, improving transparency and motivation. Progress history is logged for auditing, administrators can track aggregate metrics across the platform, and stages are configurable by destination or program type. Completion status is also updated automatically using quiz outcomes and document verification results.

11.4 Quiz and Assessment Module

Students can complete language and academic quizzes online through an intuitive interface. Agents and teachers can create and assign quizzes, and all attempts are recorded for review. Objective questions are scored automatically, while subjective answers can be evaluated manually by agents. Pass/fail outcomes are determined by configurable thresholds, students receive immediate feedback after completion, and quiz results feed directly into progress tracking.

11.5 Document Management

Students can securely upload required documents, including CVs, passports, and identity cards. Agents can verify, approve, or reject submissions efficiently, and both parties can track upload history and verification status. Document state is integrated into progress workflows, files are protected with encryption, and verification actions are logged to support compliance and traceability.

11.6 Feedback and Review System

Students can submit ratings and written feedback about agent support after interactions or quiz-related guidance. Administrators can monitor agent performance through aggregated summaries, and the agent review process supports academic feedback on student quiz responses. Historical feedback is preserved for trend analysis and continuous quality improvement.

11.7 Admin Dashboard

Administrators have access to real-time activity statistics and reports covering total users, active students, quiz completion rates, document verification status, agent performance, and ticket resolution times. The dashboard also includes destination management, product key management for contract-based registration control, and dynamic role assignment. These capabilities support data-driven decisions and continuous operational monitoring.

11.8 Secure and Reliable Platform

The platform enforces secure JWT-based authentication and strict role-based access control (ADMIN, AGENT, STUDENT, GUEST). Sensitive data, including credentials and personal documents, is protected through encryption. Daily automated backups reduce data-loss risk, while rate limiting, input sanitization, HTTPS enforcement, and session/token expiration policies improve resilience against malicious activity.

11.9 Benefits to LinkedU

For Students Students benefit from personalized guidance, clearer visibility of progress and next steps, centralized access to quizzes and documents, instant communication through chat, transparent verification and scoring outcomes, and structured ticket submission for issue resolution.

For Agents Agents benefit from a consolidated student dashboard, faster communication and document handling, improved multi-student progress oversight, integrated quiz creation and evaluation tools, structured ticket tracking, and clearer visibility of pending actions.

For Administrators Administrators gain a centralized view of users and activities, real-time performance and progress analytics, stronger oversight for decision-making, flexible user-role administration, operational control through destination/product key modules, and improved ability to detect bottlenecks.

For LinkedU as an Organization LinkedU benefits from faster onboarding and guidance processes, higher student satisfaction through transparency, lower manual effort and error rates, scalable architecture for future growth, complete audit trails for compliance, stronger market differentiation, and reduced operational costs.

11.10 Additional Outcomes from Scrum Methodology

Scrum practices are expected to enable faster bug detection and repair, higher software quality through iterative feedback integration, improved stakeholder satisfaction through regular sprint reviews, reduced delivery risk via early issue identification, greater adaptability for future feature requests, and comprehensive documentation that supports technical maintenance and user training.

12 Conclusion

The development of the LinkedU platform represents a significant step toward modernizing student guidance and academic support. The platform integrates sixteen functional modules, including authentication, user management, destination management, product key management, agent and student profiles, quiz management, quiz flow and attempts, agent review, chat, document management, progress tracking, ticket system, and student management, into a single user-friendly system.

12.1 Project Achievements

Core Objectives Met The main goal was to facilitate interaction between students, agents, and administrators, streamline the academic orientation process, and provide a reliable, secure, and scalable environment. Each module was designed to address real-world needs. Students can track progress, access guidance, take quizzes, upload documents, communicate via chat, and submit tickets. Agents can manage assigned students, create and evaluate quizzes, verify documents, respond to tickets, and update progress. Administrators can monitor performance, manage destinations and product keys, assign user roles, and generate reports.

Technical Accomplishments Technical outcomes include fifty-five or more API endpoints following RESTful conventions, JWT-based authentication with role-based access control for ADMIN, AGENT, STUDENT, and GUEST roles, a real-time chat messaging system, document upload and verification workflows, complete quiz creation/submission/evaluation/scoring capabilities, progress tracking across predefined workflow stages, and a structured ticket-handling system.

12.2 Scrum Methodology Outcomes

The adoption of Scrum methodology with continuous testing, feedback, and repair cycles produced measurable benefits: faster bug detection and resolution through daily triage, higher software quality through iterative testing after each sprint, stronger stakeholder satisfaction through regular sprint reviews, reduced project risk through early issue identification, and adaptable platform architecture for future needs.

12.3 Personal and Professional Development

Technical Skills This internship provided an opportunity to apply and strengthen full-stack development competencies using Java Spring Boot or Node.js for back end and Angular for front end, database design and management with MySQL or PostgreSQL, RESTful API design and documentation, and UML modeling for system and database architecture.

Project Management Skills Project management growth included practical implementation of Agile and Scrum practices, backlog refinement and story-point estimation, stakeholder coordination, and structured requirement gathering.

Professional Skills Professional development included coordinating with LinkedU stakeholders across multiple feedback cycles, integrating multiple modules into one cohesive platform, and ensuring both functional and non-functional requirements such as security, performance, and scalability.

12.4 Platform Readiness

The LinkedU platform is ready to serve as a robust, efficient, and scalable tool for improving student guidance and academic success.

Technical Deliverables Completed technical deliverables include a fully functional web application with all sixteen modules integrated, a production-ready database schema, complete API documentation, deployment scripts and infrastructure configuration, and automated backup and monitoring setup.

Documentation Deliverables Completed documentation deliverables include the system architecture document, database schema documentation, API reference documentation, deployment and operations guide, administrator manual, and agent/student user guides.

Training Deliverables Training deliverables include live sessions for LinkedU staff, role-specific training videos, and quick reference guides with FAQ documentation.

12.5 Final Assessment

This project achieved its stated objectives and delivered a comprehensive learning experience that bridged theoretical knowledge with practical implementation. The integration of continuous testing, feedback loops, and repair cycles throughout Scrum-based development ensured that the final product aligns with stakeholder quality expectations.

The platform delivers measurable value to all user roles. Students receive personalized guidance, clear progress visibility, and efficient communication. Agents gain stronger organization tools, faster verification workflows, and comprehensive student oversight. Administrators obtain centralized control, real-time analytics, and improved decision-making. LinkedU gains increased operational efficiency, enhanced student satisfaction, reduced manual errors, and a scalable digital platform.

12.6 Closing Remarks

The LinkedU platform transforms student guidance from a manual and fragmented process into an integrated, automated, and transparent system. Its sixteen modules work together to deliver a seamless experience from registration through quiz completion, document verification, progress tracking, and continuous agent support. The project stands as a concrete example of how structured requirements analysis, iterative development, continuous testing, and stakeholder collaboration can produce a high-quality software solution.